Scenario Task 2 — "Design Repositories for Two Different Organizations"

Company A — Startup

Create the 3 repos on GitHub

Clone all 3 repos to your machine

```
1mmun1ty@1mmun1ty MINGW64 /d/Intership Task/Week 1/Task 2 (main)
$ git clone git@github.com:1mun1ty/backend-api-.git
Cloning into 'backend-api-' ...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

1mmun1ty@1mmun1ty MINGW64 /d/Intership Task/Week 1/Task 2 (main)
$ git clone git@github.com:1mun1ty/frontend-ui.git
Cloning into 'frontend-ui' ...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

1mmun1ty@1mmun1ty MINGW64 /d/Intership Task/Week 1/Task 2 (main)
$ git clone git@github.com:1mun1ty/infra-devops.git
Cloning into 'infra-devops' ...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
```

Set up backend-api repo

```
1mmun1ty@1mmun1ty MINGW64 /d/Intership Task/Week 1/Task 2/backend-api- (main)
$ mkdir src tests docs
touch src/app.py tests/test_app.py docs/README.md .gitignore

echo "# Backend API" > README.md
echo "node_modules/\n.env\n__pycache__/" > .gitignore

git add .
git commit -m "Initial project structure"
git push origin main
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
[main 2f08c3e] Initial project structure
 5 files changed, 2 insertions(+), 1 deletion(-)
 create mode 100644 .gitignore
 create mode 100644 docs/README.md
 create mode 100644 src/app.py
 create mode 100644 tests/test_app.py
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (8/8), 581 bytes | 193.00 KiB/s, done.
Total 8 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:1mun1ty/backend-api-.git
   f2b7a3c..2f08c3e  main -> main
```

Set up frontend-ui repo

```
1mmun1ty@1mmun1ty MINGW64 /d/Intership Task/Week 1/Task 2/frontend-ui (main)
$ mkdir src public
touch src/index.html src/style.css src/main.js .gitignore

echo "# Frontend UI" > README.md
echo "node_modules/\ndist/\n.env" > .gitignore

git add .
git commit -m "Initial frontend structure"
git push origin main
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
[main e500eb0] Initial frontend structure
 5 files changed, 2 insertions(+), 1 deletion(-)
 create mode 100644 .gitignore
 create mode 100644 src/index.html
 create mode 100644 src/main.js
 create mode 100644 src/style.css
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 442 bytes | 147.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:1mun1ty/frontend-ui.git
   ed58e29..e500eb0  main -> main
```

Set up infra-devops repo

```
1mmun1ty@1mmun1ty MINGW64 /d/Intership Task/Week 1/Task 2/infra-devops (main)
$ mkdir scripts ci-cd configs
touch scripts/deploy.sh ci-cd/pipeline.yml configs/nginx.conf .gitignore

echo "# Infra & DevOps" > README.md
echo "*.tfstate\n*.tfvars\n.env\nsecrets/" > .gitignore

git add .
git commit -m "Initial infra structure"
git push origin main
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
[main 5f2dda0] Initial infra structure
 5 files changed, 2 insertions(+), 1 deletion(-)
 create mode 100644 .gitignore
 create mode 100644 ci-cd/pipeline.yml
 create mode 100644 configs/nginx.conf
 create mode 100644 scripts/deploy.sh
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (8/8), 595 bytes | 198.00 KiB/s, done.
Total 8 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:1mun1ty/infra-devops.git
   76ad55f..5f2dda0  main -> main
```

Branching strategy (simple)

```
1mmun1ty@1mmun1ty MINGW64 /d/Intership Task/Week 1/Task 2/backend-api- (main)
$ git checkout -b feature/user-login
git add .
git commit -m "Add user login endpoint"
git push origin feature/user-login

git checkout main
git merge feature/user-login
git push origin main
git branch -d feature/user-login
Switched to a new branch 'feature/user-login'
On branch feature/user-login
nothing to commit, working tree clean
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/user-login' on GitHub by visiting:
remote:      https://github.com/1mun1ty/backend-api-/pull/new/feature/user-login
remote:
To github.com:1mun1ty/backend-api-.git
 * [new branch]      feature/user-login -> feature/user-login
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
Already up to date.
Everything up-to-date
Deleted branch feature/user-login (was 2f08c3e).
```

Company B — Medium Org

Create all microservice repos

Service repos

svc-notifications Public

Updated 1 minute ago

---

svc-payments Public

Updated 1 minute ago

---

svc-orders Public

Updated 3 minutes ago

---

svc-users Public

Updated 4 minutes ago

---

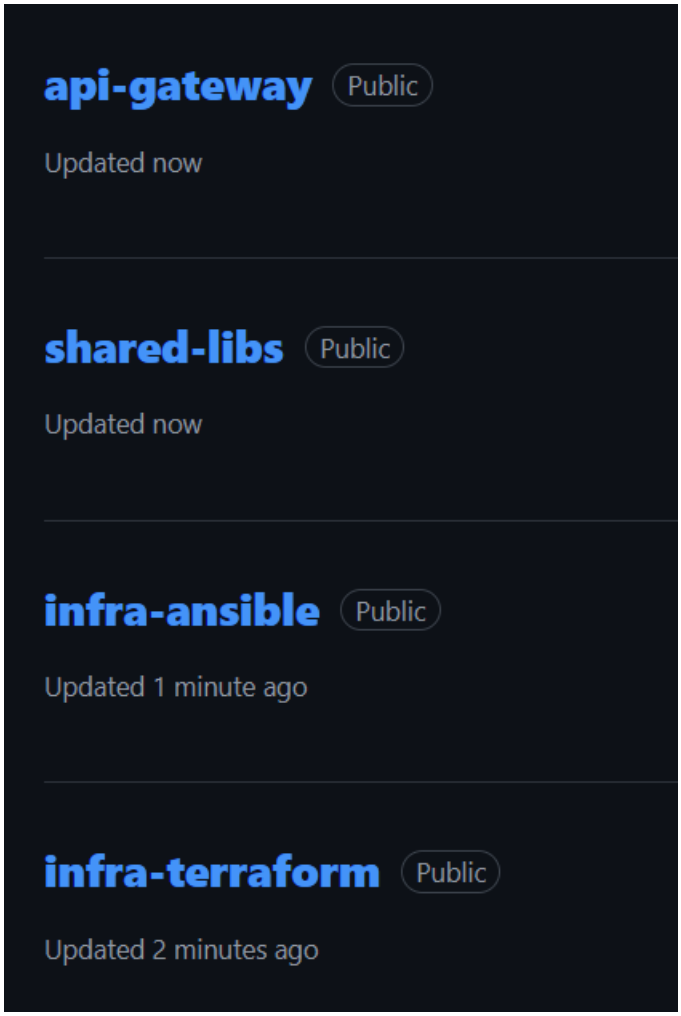
svc-auth Public

Updated 5 minutes ago

Create each repo on GitHub, then clone them all locally.

```
1mmun1ty@1mmun1ty MINGW64 /d/Intership Task/Week 1/Task 2 (main)
$ git clone git@github.com:1mun1ty/svc-notifications.git
Cloning into 'svc-notifications'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

1mmun1ty@1mmun1ty MINGW64 /d/Intership Task/Week 1/Task 2 (main)
$ git clone git@github.com:1mun1ty/svc-payments.git
Cloning into 'svc-payments'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

1mmun1ty@1mmun1ty MINGW64 /d/Intership Task/Week 1/Task 2 (main)
$ git clone git@github.com:1mun1ty/svc-orders.git
Cloning into 'svc-orders'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

1mmun1ty@1mmun1ty MINGW64 /d/Intership Task/Week 1/Task 2 (main)
$ git clone git@github.com:1mun1ty/svc-users.git
Cloning into 'svc-users'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

1mmun1ty@1mmun1ty MINGW64 /d/Intership Task/Week 1/Task 2 (main)
$ git clone git@github.com:1mun1ty/svc-auth.git
Cloning into 'svc-auth'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

1mmun1ty@1mmun1ty MINGW64 /d/Intership Task/Week 1/Task 2 (main)
$
```

Shared repos



Create each repo on GitHub, then clone them all locally.

```
1mmun1ty@1mmun1ty MINGW64 /d/Intership Task/Week 1/Task 2 (main)
$ git clone git@github.com:1mun1ty/api-gateway.git
Cloning into 'api-gateway'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

1mmun1ty@1mmun1ty MINGW64 /d/Intership Task/Week 1/Task 2 (main)
$ git clone git@github.com:1mun1ty/shared-libs.git
Cloning into 'shared-libs'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

1mmun1ty@1mmun1ty MINGW64 /d/Intership Task/Week 1/Task 2 (main)
$ git clone git@github.com:1mun1ty/infra-ansible.git
Cloning into 'infra-ansible'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.

1mmun1ty@1mmun1ty MINGW64 /d/Intership Task/Week 1/Task 2 (main)
$ git clone git@github.com:1mun1ty/infra-terraform.git
Cloning into 'infra-terraform'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
```

Set up a microservice repo (svc-auth)

```
1mmun1ty@1mmun1ty MINGW64 /d/Intership Task/Week 1/Task 2/svc-auth (main)
$ mkdir src tests docs .github .github/workflows

touch src/main.py
touch tests/test_auth.py
touch docs/api.md
touch Dockerfile
touch docker-compose.yml
touch requirements.txt
touch .env.example
touch .gitignore

echo "# Auth Service" > README.md
echo ".env\n__pycache__/\n*.pyc\n.DS_Store" > .gitignore

git add .
git commit -m "Initial auth service structure"
git push origin main
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
[main 743ee93] Initial auth service structure
 9 files changed, 2 insertions(+), 1 deletion(-)
 create mode 100644 .env.example
 create mode 100644 .gitignore
 create mode 100644 Dockerfile
 create mode 100644 docker-compose.yml
 create mode 100644 docs/api.md
 create mode 100644 requirements.txt
 create mode 100644 src/main.py
 create mode 100644 tests/test_auth.py
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (8/8), 668 bytes | 111.00 KiB/s, done.
Total 8 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:1mun1ty/svc-auth.git
   2f721d0..743ee93  main -> main
```

Set up infra-terraform repo

```
1mmun1ty@1mmun1ty MINGW64 /d/Intership Task/Week 1/Task 2/infra-terraform (main)
$ mkdir environments modules
mkdir environments/dev environments/staging environments/prod
mkdir modules/vpc modules/eks modules/rds

touch environments/dev/main.tf environments/dev/variables.tf
touch environments/prod/main.tf environments/prod/variables.tf
touch modules/vpc/main.tf modules/vpc/outputs.tf
touch .gitignore

echo "*.tfstate\n*.tfstate.backup\n.terraform/\n*.tfvars\nsecrets/" > .gitignore
echo "# Infrastructure - Terraform" > README.md

git add .
git commit -m "Initial Terraform structure"
git push origin main
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it
[main b379134] Initial Terraform structure
 8 files changed, 2 insertions(+), 1 deletion(-)
 create mode 100644 .gitignore
 create mode 100644 environments/dev/main.tf
 create mode 100644 environments/dev/variables.tf
 create mode 100644 environments/prod/main.tf
 create mode 100644 environments/prod/variables.tf
 create mode 100644 modules/vpc/main.tf
 create mode 100644 modules/vpc/outputs.tf
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (9/9), 662 bytes | 165.00 KiB/s, done.
Total 9 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:1mun1ty/infra-terraform.git
   a3a95eb..b379134  main -> main
```

Branching strategy (GitFlow)

```
1mmun1ty@1mmun1ty MINGW64 /d/Intership Task/Week 1/Task 2 (main)
$ cd svc-auth

git checkout main
git checkout -b develop
git push origin develop

git checkout -b feature/jwt-tokens
git add .
git commit -m "Add JWT token generation"
git push origin feature/jwt-tokens
Already on 'main'
Your branch is up to date with 'origin/main'.
Switched to a new branch 'develop'
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:      https://github.com/1mun1ty/svc-auth/pull/new/develop
remote:
To github.com:1mun1ty/svc-auth.git
 * [new branch]      develop -> develop
Switched to a new branch 'feature/jwt-tokens'
On branch feature/jwt-tokens
nothing to commit, working tree clean
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/jwt-tokens' on GitHub by visiting:
remote:      https://github.com/1mun1ty/svc-auth/pull/new/feature/jwt-tokens
remote:
To github.com:1mun1ty/svc-auth.git
 * [new branch]      feature/jwt-tokens -> feature/jwt-tokens
```

Add CI/CD pipeline to a service

```
1mmun1ty@1mmun1ty MINGW64 /d/Intership Task/Week 1/Task 2 (main)
$ cd svc-auth

cat > .github/workflows/ci.yml << 'EOF'
name: CI Pipeline

on:
  push:
    branches: [main, develop]
  pull_request:
    branches: [main, develop]

jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3
      - name: Set up Python
        uses: actions/setup-python@v4
        with:
          python-version: '3.11'
      - name: Install dependencies
        run: pip install -r requirements.txt
      - name: Run tests
        run: pytest tests/

  build:
    needs: test
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3
      - name: Build Docker image
        run: docker build -t svc-auth:latest .
EOF

git add .github/workflows/ci.yml
git commit -m "Add CI pipeline"
git push origin feature/jwt-tokens
warning: in the working copy of '.github/workflows/ci.yml', LF will be replaced by CRLF the next time Git touches it
[feature/jwt-tokens fb8b062] Add CI pipeline
 1 file changed, 29 insertions(+)
 create mode 100644 .github/workflows/ci.yml
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 637 bytes | 318.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:1mun1ty/svc-auth.git
   743ee93..fb8b062  feature/jwt-tokens -> feature/jwt-tokens
```